H524 Introduction to Biostatistics
Week 7 Lab: Nonparametric Methods
Fall 2025

Format: Online
Tools Required: R, RStudio
Estimated Time: 90 minutes

1 Learning Objectives

By the end of this lab, you will be able to:

1. Check normality assumptions using Shapiro-Wilk test
2. Choose between parametric and nonparametric tests
3. Perform sign tests in R
4. Use Wilcoxon signed-rank test for one-sample and paired data
5. Conduct Wilcoxon rank-sum tests for two groups
6. Perform Kruskal-Wallis tests for three or more groups
7. Interpret nonparametric test output
8. Compare parametric and nonparametric results

2 Introduction: When to Use Nonparametric Tests

Parametric tests (t-tests, ANOVA) assume data are normally distributed. When this assumption is violated, use nonparametric tests instead.

Use nonparametric tests when:

- Data are significantly skewed
- Extreme outliers are present
- Small sample size (n ¡ 20)
- Ordinal data (pain scales, ratings)
- Shapiro-Wilk test p-value ¡ 0.05

Quick Reference:

Scenario	Normal Data	Not Normal
One sample	t-test	Sign or Wilcoxon
Paired data	Paired t-test	Sign or Wilcoxon
Two groups	Two-sample t-test	Wilcoxon rank-sum
3+ groups	ANOVA	Kruskal-Wallis

3 Part 1: Checking Normality

Always check normality before choosing your test.

3.1 Exercise 1: Shapiro-Wilk Test

The Shapiro-Wilk test checks if data are normally distributed.

- H_0: Data are normal
- If $p \geq 0.05$: Use parametric test
- If $p < 0.05$: Use nonparametric test

```r
# Generate example data
set.seed(524)
normal_data <- rnorm(30, mean=100, sd=15)
skewed_data <- rexp(30, rate=0.02)

# Check normality
shapiro.test(normal_data)
shapiro.test(skewed_data)

# Visual check (optional)
hist(normal_data, main="Normal Data", col="lightblue")
hist(skewed_data, main="Skewed Data", col="lightcoral")

# Q-Q plots
qqnorm(normal_data, main="Normal Data Q-Q Plot")
qqline(normal_data, col="red")

qqnorm(skewed_data, main="Skewed Data Q-Q Plot")
qqline(skewed_data, col="red")
```

Questions:

1. What is the p-value for `normal_data`?
2. What is the p-value for `skewed_data`?
3. For `normal_data`, should you use parametric or nonparametric tests?
4. For `skewed_data`, should you use parametric or nonparametric tests?

4 Part 2: Sign Test

The sign test only uses the direction (+/-) of differences.

4.1 Exercise 2: Sign Test for One Sample

Question: Is median recovery time different from 7 days?

```
# Recovery times (days)
recovery <- c(8, 5, 7, 10, 4, 9, 3, 12, 6, 11, 7)

# Hypothesized median
med0 <- 7

# Calculate differences
diff <- recovery - med0

# Count signs (ignoring zeros)
n_positive <- sum(diff > 0)
n_total <- sum(diff != 0)

cat("Positive differences:", n_positive, "\n")
cat("Total non-zero:", n_total, "\n")

# Sign test
binom.test(n_positive, n_total, p=0.5)
```

Questions:

1. How many positive differences are there?
2. How many total non-zero differences?
3. What is the p-value?
4. Is the median significantly different from 7 days?

4.2 Exercise 3: Sign Test for Paired Data

Question: Does treatment reduce pain scores?

```
# Pain scores (1-10 scale)
before <- c(7, 8, 6, 9, 5, 8, 7, 6)
after <- c(5, 6, 4, 7, 4, 6, 5, 5)

# Calculate differences
diff <- before - after

# Count signs
n_improved <- sum(diff > 0)
n_total <- sum(diff != 0)

cat("Number improved:", n_improved, "\n")
cat("Number total:", n_total, "\n")

# Sign test
binom.test(n_improved, n_total, p=0.5)
```

```
# Summary
mean(before)
mean(after)
mean(diff)
```

Questions:

1. How many patients improved?
2. What is the p-value?
3. Does the treatment reduce pain?
4. What is the average reduction in pain score?

5 Part 3: Wilcoxon Signed-Rank Test (Paired Data)

Use for paired data when normality assumption is violated.

5.1 Exercise 4: Blood Pressure Study

Research Question: Does medication reduce blood pressure?

```
# Blood pressure before and after medication
bp_before <- c(145, 138, 150, 142, 156, 148, 140, 152)
bp_after <- c(138, 135, 142, 140, 148, 145, 138, 146)

# Check normality of differences
diff <- bp_before - bp_after
shapiro.test(diff)

# Wilcoxon signed-rank test
wilcox.test(bp_before, bp_after, paired=TRUE)

# Summary statistics
cat("Mean BP before:", mean(bp_before), "\n")
cat("Mean BP after:", mean(bp_after), "\n")
cat("Mean reduction:", mean(diff), "\n")
```

Questions:

1. What is the Shapiro-Wilk p-value for the differences? Is normality OK?
2. What is the Wilcoxon test p-value?
3. Is the difference significant at $\alpha = 0.05$?
4. Does the medication reduce blood pressure?

5. What is the mean reduction in blood pressure?

5.2 Exercise 5: Physical Therapy Study

Question: Does physical therapy reduce pain scores?

```
# Pain scores (0-10 scale) before and after 8 weeks PT
pain_before <- c(7, 8, 6, 9, 7, 8, 6, 7, 8, 9)
pain_after <- c(5, 6, 5, 7, 6, 7, 4, 6, 7, 8)

# Check normality
diff <- pain_before - pain_after
shapiro.test(diff)

# Wilcoxon test
wilcox.test(pain_before, pain_after, paired=TRUE)

# Medians
median(pain_before)
median(pain_after)
median(diff)
```

Questions:

1. Are the differences normally distributed?
2. What is the Wilcoxon p-value?
3. Does PT significantly reduce pain?
4. What is the median reduction in pain score?

6 Part 4: Wilcoxon Rank-Sum Test (Two Independent Groups)

Use for comparing two independent groups when normality is violated.

6.1 Exercise 6: Treatment Comparison

Research Question: Do two treatments differ in recovery time?

```
# Recovery times (days)
treatment_A <- c(23, 31, 25, 28, 30, 27, 29)
treatment_B <- c(18, 22, 20, 24, 19, 21, 23)

# Check normality
```

```
shapiro.test(treatment_A)
shapiro.test(treatment_B)

# Visualize
boxplot(treatment_A, treatment_B,
        names=c("Treatment A", "Treatment B"),
        ylab="Recovery Time (days)",
        main="Recovery Time Comparison",
        col=c("lightblue", "lightcoral"))

# Wilcoxon rank-sum test
wilcox.test(treatment_A, treatment_B)

# Summary statistics
cat("Median Treatment A:", median(treatment_A), "days\n")
cat("Median Treatment B:", median(treatment_B), "days\n")
```

Questions:

1. Are the data normal for Treatment A? Treatment B?
2. What is the Wilcoxon rank-sum p-value?
3. Do the treatments differ significantly at $\alpha = 0.05$?
4. Which treatment has shorter recovery times?
5. What are the median recovery times for each treatment?

6.2 Exercise 7: Drug Effectiveness Study

Question: Do two drugs differ in effectiveness?

```
# Symptom relief scores (higher = better)
drug_X <- c(45, 52, 48, 55, 50, 47, 53)
drug_Y <- c(38, 42, 40, 45, 39, 43, 41)

# Visualize
boxplot(drug_X, drug_Y,
        names=c("Drug X", "Drug Y"),
        ylab="Symptom Relief Score",
        col=c("lightgreen", "lightyellow"))

# Wilcoxon test
wilcox.test(drug_X, drug_Y)

# Summary
mean(drug_X)
mean(drug_Y)
median(drug_X)
```

```
median(drug_Y)
```

Questions:

1. What is the Wilcoxon p-value?
2. Which drug is more effective?
3. What is the difference in average scores between the drugs?

7 Part 5: Kruskal-Wallis Test (Three or More Groups)

Use for comparing 3+ independent groups when normality is violated.

7.1 Exercise 8: Pain Relief Comparison

Research Question: Do three pain medications differ in effectiveness?

```
# Pain relief scores (0-10 scale, higher = better)
drug_A <- c(3, 4, 5, 4, 3, 5, 4)
drug_B <- c(5, 6, 7, 6, 5, 7, 6)
drug_C <- c(7, 8, 9, 8, 7, 9, 8)

# Combine data
pain_scores <- c(drug_A, drug_B, drug_C)
drug <- factor(rep(c("A", "B", "C"), each=7))

# Check normality for each group
shapiro.test(drug_A)
shapiro.test(drug_B)
shapiro.test(drug_C)

# Visualize
boxplot(pain_scores ~ drug,
        xlab="Drug",
        ylab="Pain Relief Score",
        main="Pain Relief by Drug",
        col=c("lightblue", "lightgreen", "lightyellow"))

# Kruskal-Wallis test
kruskal.test(pain_scores ~ drug)

# Post-hoc pairwise comparisons
pairwise.wilcox.test(pain_scores, drug, p.adjust.method="bonferroni")

# Summary statistics
tapply(pain_scores, drug, median)
```

Questions:

1. What is the Kruskal-Wallis p-value?
2. Do the drugs differ significantly overall at $\alpha = 0.05$?
3. According to the post-hoc tests, which specific pairs of drugs differ significantly?
4. Which drug appears most effective based on median scores?
5. Which drug appears least effective?

8 Part 6: Comparing Parametric vs Nonparametric

8.1 Exercise 9: When Both Tests Apply

Sometimes data are borderline normal. Let's compare both approaches.

```
# Blood pressure data (from Exercise 4)
bp_before <- c(145, 138, 150, 142, 156, 148, 140, 152)
bp_after <- c(138, 135, 142, 140, 148, 145, 138, 146)

# Check normality
diff <- bp_before - bp_after
shapiro.test(diff)

# Parametric: Paired t-test
t_result <- t.test(bp_before, bp_after, paired=TRUE)
cat("Paired t-test p-value:", t_result$p.value, "\n")

# Nonparametric: Wilcoxon
w_result <- wilcox.test(bp_before, bp_after, paired=TRUE)
cat("Wilcoxon p-value:", w_result$p.value, "\n")

# Compare
cat("\nBoth tests significant? Compare p-values to 0.05\n")
```

Questions:

1. Is the normality assumption met ($p \geq 0.05$)?
2. Which test gives a smaller p-value?
3. Do both tests reach the same conclusion (significant or not)?
4. When data are borderline normal, which test is safer to use?

9 Practice Problems

9.1 Problem 1: Hospital Stay Duration

Compare median hospital stay between two surgical procedures.

```
# Hospital stay (days)
procedure_1 <- c(3, 4, 5, 3, 6, 4, 7, 5, 4, 12)
procedure_2 <- c(5, 6, 8, 7, 9, 6, 8, 10, 7, 15)

# Step 1: Visualize
boxplot(procedure_1, procedure_2,
        names=c("Procedure 1", "Procedure 2"),
        main="Hospital Stay Comparison",
        ylab="Days",
        col=c("lightblue", "lightcoral"))

# Step 2: Check normality
shapiro.test(procedure_1)
shapiro.test(procedure_2)

# Step 3: Wilcoxon rank-sum test
wilcox.test(procedure_1, procedure_2)

# Step 4: Summary statistics
cat("Median Procedure 1:", median(procedure_1), "days\n")
cat("Median Procedure 2:", median(procedure_2), "days\n")
```

Questions:

1. Are the data normal for each procedure?
2. What is the Wilcoxon rank-sum p-value?
3. Do the procedures differ significantly in hospital stay?
4. Which procedure has shorter hospital stays?
5. Why is Wilcoxon appropriate here? (Hint: check for outliers)

9.2 Problem 2: Diet Program Comparison

Compare weight loss across four diet programs.

```
# Weight loss (kg) after 3 months
diet_A <- c(2.1, 3.2, 2.5, 3.0, 2.8, 2.6)
diet_B <- c(3.5, 4.2, 3.8, 4.0, 3.9, 3.7)
diet_C <- c(4.5, 5.1, 4.8, 5.3, 4.9, 5.0)
diet_D <- c(2.8, 3.0, 2.9, 3.1, 2.7, 2.9)
```

```
# Combine data
weight_loss <- c(diet_A, diet_B, diet_C, diet_D)
diet <- factor(rep(c("A", "B", "C", "D"), each=6))

# Visualize
boxplot(weight_loss ~ diet,
        main="Weight Loss by Diet Program",
        xlab="Diet",
        ylab="Weight Loss (kg)",
        col=c("lightblue", "lightgreen", "lightyellow", "lightcoral"))

# Kruskal-Wallis test
kruskal.test(weight_loss ~ diet)

# Post-hoc tests
pairwise.wilcox.test(weight_loss, diet, p.adjust.method="bonferroni")

# Summary statistics
tapply(weight_loss, diet, median)
tapply(weight_loss, diet, mean)
```

Questions:

1. What is the Kruskal-Wallis p-value?
2. Do the diets differ significantly overall?
3. Which specific pairs of diets differ significantly? (Check post-hoc p-values ¡ 0.05)
4. Which diet appears most effective for weight loss?
5. What is the median weight loss for each diet?

9.3 Problem 3: Satisfaction Ratings (Ordinal Data)

Compare customer satisfaction ratings across three stores.

```
# Satisfaction ratings (1-5 scale, ordinal)
store_A <- c(3, 4, 3, 5, 4, 3, 4, 3, 4, 3)
store_B <- c(4, 5, 4, 5, 5, 4, 5, 4, 5, 4)
store_C <- c(2, 3, 2, 4, 3, 2, 3, 2, 3, 3)

# Combine
satisfaction <- c(store_A, store_B, store_C)
store <- factor(rep(c("A", "B", "C"), each=10))

# Visualize
boxplot(satisfaction ~ store,
        main="Satisfaction by Store",
        xlab="Store",
```

```
        ylab="Satisfaction Rating (1-5)",
        col=c("lightblue", "lightgreen", "lightyellow"))

# Kruskal-Wallis (appropriate for ordinal data!)
kruskal.test(satisfaction ~ store)

# Post-hoc
pairwise.wilcox.test(satisfaction, store, p.adjust.method="bonferroni")

# Medians
tapply(satisfaction, store, median)
```

Questions:

1. Why is nonparametric appropriate here?
2. What is the Kruskal-Wallis p-value?
3. Do the stores differ in customer satisfaction?
4. Which store has the highest satisfaction?
5. Which pairs of stores differ significantly?

10 Decision Guide Summary

Step 1: Check normality

- Use `shapiro.test()`
- If $p \geq 0.05$: Normal OK $\rightarrow$ Use parametric
- If $p < 0.05$: Not normal $\rightarrow$ Use nonparametric

Step 2: Choose test

- **One sample or paired:** Sign test or `wilcox.test()`
- **Two groups:** `wilcox.test(group1, group2)`
- **3+ groups:** `kruskal.test(outcome ~ group)`
- **Post-hoc:** `pairwise.wilcox.test()`

Step 3: Interpret

- $p < 0.05$: Significant difference
- $p \geq 0.05$: No significant difference

11 Key Takeaways

1. **Always check normality first** with Shapiro-Wilk test
2. **Nonparametric tests are robust** to outliers and skewness
3. **Wilcoxon tests** are the nonparametric versions of t-tests
4. **Kruskal-Wallis** is the nonparametric version of ANOVA
5. **When in doubt**, use nonparametric (safer choice)
6. **For ordinal data**, ALWAYS use nonparametric
7. **Follow up Kruskal-Wallis** with post-hoc tests if significant